

Linear Algebra

Past Paper — Full Solutions

BS-II (1st Semester) · Final Test for Repeaters · Session 2025

SUBJECT Linear Algebra Max Marks: 80 — Attempt any 3	COMPOSED BY Nadir Ali Khan BS-IV Mathematics	DATE 05-06-2025 9:30 – 11:30 AM
---	--	---

Table of Contents

Q.NO.1 — Concept, Purpose and Applications of Matrices	2
Q.NO.2(a) — Nilpotent, Hermitian, and Involuntary Matrices	4
Q.NO.2(b) — Properties of Complex Conjugate	6
Q.NO.3 — Normal Form and Matrices P, Q	7
Q.NO.4(a) — Echelon and Reduced Echelon Matrix	10
Q.NO.4(b) — Rank and Inverse of Matrix	11
Q.NO.5 — Row and Column Operations, Reduce Matrices.....	13
Quick Reference — Linear Algebra.....	15

Q.NO.1 — Concept, Purpose and Applications of Matrices

Q.NO.1

What is the concept of a Matrix and what is the purpose and applications of matrices? Justify through examples.

Concept of a Matrix

Definition — Matrix

A **matrix** is a rectangular array of numbers (real or complex) arranged in **rows** and **columns**, enclosed in square brackets. A matrix with m rows and n columns is called an $m \times n$ matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$$

The element in the i -th row and j -th column is denoted a_{ij} .

Types of Matrices

- **Square Matrix:** $m = n$. Example: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
- **Zero (Null) Matrix:** All entries are 0. $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
- **Identity Matrix:** Square matrix with 1s on diagonal, 0s elsewhere. $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- **Diagonal Matrix:** Non-zero entries only on main diagonal.
- **Symmetric:** $A^t = A$. Example: $\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$
- **Skew-Symmetric:** $A^t = -A$, diagonal entries = 0.
- **Upper/Lower Triangular:** Zeros below/above diagonal.
- **Singular:** $\det(A) = 0$ (no inverse). **Non-singular:** $\det(A) \neq 0$.

Purpose of Matrices

- Compact representation of data and linear transformations
- Efficient solution of systems of linear equations using row operations
- Foundation of linear algebra and higher mathematics

Applications of Matrices

Field	Application
Engineering	Structural analysis, circuit equations, control systems
Computer Graphics	Rotation, scaling, translation of 2D/3D objects
Economics	Input-output models (Leontief model)
Statistics	Covariance matrices, regression, data analysis
Physics	Quantum mechanics, tensor equations
Cryptography	Encoding/decoding messages (Hill cipher)
Network Theory	Adjacency matrices for graphs

Example — Solving System of Equations:

$$\begin{cases} 2x + y = 5 \\ x - y = 1 \end{cases} \implies \underbrace{\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 5 \\ 1 \end{bmatrix}}_{\mathbf{b}} \implies \mathbf{x} = A^{-1}\mathbf{b}$$

$$A^{-1} = \frac{1}{-3} \begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix}, \text{ so } x = 2, y = 1.$$

Q.NO.2(a) — Nilpotent, Hermitian, and Involuntary Matrices

Q.NO.2(a)

What do you know about the Nilpotent matrix, Hermitian matrix and Involuntary matrix? Prove through examples.

Nilpotent Matrix

Definition — Nilpotent Matrix

A square matrix A is called **nilpotent** if there exists a positive integer k such that:

$$A^k = O \quad (\text{zero matrix})$$

The smallest such k is called the **index** of nilpotency.

Proof

Example: Let $A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

So $A^3 = O$. **Index of nilpotency = 3.** ✓

Properties: All eigenvalues of a nilpotent matrix are 0. Therefore $\det(A) = 0$ and $\text{tr}(A) = 0$.

Hermitian Matrix

Definition — Hermitian Matrix

A complex square matrix A is **Hermitian** if it equals its own conjugate transpose:

$$A = A^* \quad \text{where } A^* = \overline{A}^t = \overline{(A^t)}$$

Proof

Example: Let $A = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}$

Conjugate: $\bar{A} = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}$

Transpose of conjugate: $\bar{A}^t = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix} = A \quad \checkmark$

So $A^* = A$, hence A is **Hermitian**.

Properties: Diagonal elements of a Hermitian matrix are always **real**. All eigenvalues are real.

Involuntary Matrix**Definition — Involuntary Matrix**

A square matrix A is **involuntary** if:

$$A^2 = I \quad (\text{identity matrix})$$

i.e., A is its own inverse: $A^{-1} = A$.

Proof

Example: Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \quad \checkmark$$

So A is **involuntary**.

Properties: Eigenvalues of an involuntary matrix are ± 1 . Every involuntary matrix is non-singular with $\det(A) = \pm 1$.

Q.NO.2(b) — Properties of Complex Conjugate

Q.NO.2(b)

Show that for any matrices A, B over $\mathbb{C}$ and $k \in \mathbb{C}$:

(i) $\overline{A+B} = \overline{A} + \overline{B}$ (ii) $\overline{(A)^t} = (A^t)$ i.e., $(\overline{A})^t = \overline{A^t}$

(i) **Proof:** $\overline{A+B} = \overline{A} + \overline{B}$

Proof

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ matrices over $\mathbb{C}$.

Then $A+B = [a_{ij} + b_{ij}]$.

Taking the complex conjugate entry-wise:

$$\overline{A+B} = [\overline{a_{ij} + b_{ij}}]$$

By the property of complex conjugates ($\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$):

$$= [\overline{a_{ij}} + \overline{b_{ij}}] = [\overline{a_{ij}}] + [\overline{b_{ij}}] = \overline{A} + \overline{B}$$

$$\therefore \overline{A+B} = \overline{A} + \overline{B} \quad \blacksquare$$

(ii) **Proof:** $(\overline{A})^t = \overline{(A^t)}$

Proof

Let $A = [a_{ij}]$ be an $m \times n$ matrix over $\mathbb{C}$.

Left side: $\overline{A} = [\overline{a_{ij}}]$, so $(\overline{A})^t = [\overline{a_{ji}}]$

Right side: $A^t = [a_{ji}]$, so $\overline{A^t} = [\overline{a_{ji}}]$

Since both sides equal $[\overline{a_{ji}}]$:

$$(\overline{A})^t = \overline{A^t} \quad \blacksquare$$

This means the operations of **transposition** and **complex conjugation** commute.

Q.NO.3 — Normal Form and Matrices P, Q

Q.NO.3

Reduce the matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$ to normal form. Find nonsingular matrices P and Q such that PAQ is the normal form.

Solution**Step 1 — Find the Rank.**

We compute $\det(A)$:

$$\det(A) = 1(2(-1) - 3(-1)) - 1(1(-1) - 3(0)) + 2(1(-1) - 2(0)) = 1(1) - 1(-1) + 2(-1) = 1 + 1 - 2 = 0$$

Since $\det(A) = 0$, $\text{rank} < 3$. We will show $\text{rank} = 2$.

Step 2 — Row Operations (tracking P).

We apply row operations to A and track the same operations on P (starting as I_3):

Initial:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_2 \leftarrow R_2 - R_1$:

$$A \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}, \quad P \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_3 \leftarrow R_3 + R_2$:

$$A \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad P \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$R_1 \leftarrow R_1 - R_2$:

$$A \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad P \rightarrow \begin{bmatrix} 2 & -1 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

Step 3 — Column Operations (tracking Q).

Starting from $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $Q = I_3$:

$C_3 \leftarrow C_3 - C_1$:

$$A \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad Q \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$C_3 \leftarrow C_3 - C_2$:

$$A \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad Q \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 4 — Result.

$$P = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Normal Form} = PAQ \cong \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank of $A = 2$.

Proof**Verification** — Compute PA :

$$PA = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Then $(PA)Q$:

$$(PA)Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \checkmark$$

Q.NO.4(a) — Echelon and Reduced Echelon Matrix

Q.NO.4(a)

Define the Echelon matrix and Reduced Echelon matrix with solution through any matrix of each.

Echelon Form (Row Echelon Form)

Definition — Echelon Matrix

A matrix is in **Row Echelon Form (REF)** if:

1. All zero rows are at the **bottom**
2. The leading entry (pivot) of each nonzero row is strictly to the **right** of the pivot in the row above
3. All entries **below** each pivot are zero

Solution — Example — Echelon Form

Reduce $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$ to Echelon Form.
 $R_2 \leftarrow R_2 - 2R_1$; $R_3 \leftarrow R_3 - 3R_1$:

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_3 \leftarrow R_3 - R_2$:

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \leftarrow \text{Echelon Form}$$

Reduced Row Echelon Form (RREF)

Definition — Reduced Echelon Matrix

A matrix is in **Reduced Row Echelon Form (RREF)** if it satisfies all conditions of REF **plus**:

1. Each pivot is 1
2. All entries **above and below** each pivot are zero

Solution — Example — RREF

Continuing from the echelon form above:

$$R_1 \leftarrow R_1 - 3R_2:$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \leftarrow \text{Reduced Echelon Form (RREF)}$$

Pivots are 1, and all entries above/below each pivot are zero. ✓

Q.NO.4(b) — Rank and Inverse of Matrix**Q.NO.4(b)**

Find the inverse and rank of $A = \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$

Solution**Step 1 — Compute $\det(A)$.**

$$\begin{aligned}
 \det(A) &= 5 \begin{vmatrix} 5 & 6 \\ -5 & -3 \end{vmatrix} - 9 \begin{vmatrix} -3 & 6 \\ -1 & -3 \end{vmatrix} + 3 \begin{vmatrix} -3 & 5 \\ -1 & -5 \end{vmatrix} \\
 &= 5(5 \cdot (-3) - 6 \cdot (-5)) - 9((-3)(-3) - 6(-1)) + 3((-3)(-5) - 5(-1)) \\
 &= 5(-15 + 30) - 9(9 + 6) + 3(15 + 5) \\
 &= 5(15) - 9(15) + 3(20) \\
 &= 75 - 135 + 60 = 0
 \end{aligned}$$

Since $\det(A) = 0$, the matrix A is **singular**. Therefore the **inverse does not exist**.

Step 2 — Find the Rank by Row Reduction.Swap $R_1 \leftrightarrow R_3$:

$$\rightarrow \begin{bmatrix} -1 & -5 & -3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix}$$

 $R_2 \leftarrow R_2 - 3R_1$:

$$\rightarrow \begin{bmatrix} -1 & -5 & -3 \\ 0 & 20 & 15 \\ 5 & 9 & 3 \end{bmatrix}$$

 $R_3 \leftarrow R_3 + 5R_1$:

$$\rightarrow \begin{bmatrix} -1 & -5 & -3 \\ 0 & 20 & 15 \\ 0 & -16 & -12 \end{bmatrix}$$

 $R_3 \leftarrow R_3 + \frac{4}{5}R_2$ (since $-16 + \frac{4}{5} \cdot 20 = 0$):

$$\rightarrow \begin{bmatrix} -1 & -5 & -3 \\ 0 & 20 & 15 \\ 0 & 0 & 0 \end{bmatrix}$$

Two nonzero rows $\Rightarrow \mathbf{Rank}(A) = 2$.

Q.NO.5 — Row and Column Operations, Reduce Matrices

Q.NO.5

Define Row and Column operations with example of each. Reduce the given matrices to the specified forms.

Row Operations

Definition — Elementary Row Operations

Three types of elementary row operations:

1. **Row Swap:** $R_i \leftrightarrow R_j$ — interchange two rows
2. **Row Scaling:** $R_i \leftarrow k R_i$ — multiply a row by nonzero scalar k
3. **Row Addition:** $R_i \leftarrow R_i + k R_j$ — add k times row j to row i

Column Operations

Definition — Elementary Column Operations

Three analogous column operations:

1. **Column Swap:** $C_i \leftrightarrow C_j$
2. **Column Scaling:** $C_i \leftarrow k C_i$
3. **Column Addition:** $C_i \leftarrow C_i + k C_j$

Matrix 1 — Reduce to Reduced Echelon Form

$$B = \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{bmatrix}$$

Solution — RREF Solution

$$R_3 \leftarrow R_3 - 2R_1:$$

$$\rightarrow \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & -11 & 5 & -3 \\ 4 & 1 & 1 & 5 \end{bmatrix}$$

$$R_4 \leftarrow R_4 - 4R_1:$$

$$\rightarrow \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & -11 & 5 & -3 \\ 0 & -11 & 5 & -3 \end{bmatrix}$$

$$R_3 \leftarrow R_3 + R_2:$$

$$\rightarrow \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & -11 & 5 & -3 \end{bmatrix}$$

$$R_4 \leftarrow R_4 + R_2:$$

$$\rightarrow \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \leftarrow \frac{1}{11}R_2:$$

$$\rightarrow \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -5/11 & 3/11 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2:$$

$$\text{RREF} = \begin{bmatrix} 1 & 0 & 4/11 & 13/11 \\ 0 & 1 & -5/11 & 3/11 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank = 2 (two nonzero rows).

Matrix 2 — Reduce to Echelon Form

$$C = \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 2 & 4 & 1 & -2 & 3 \\ 3 & 6 & 2 & -6 & 3 \end{bmatrix}$$

Solution — Echelon Form Solution

$$R_2 \leftarrow R_2 - 2R_1:$$

$$\rightarrow \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 3 & -6 & 1 \\ 3 & 6 & 2 & -6 & 3 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 3R_1:$$

$$\rightarrow \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 3 & -6 & 1 \\ 0 & 0 & 5 & -12 & 0 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - \frac{5}{3}R_2 \quad (\text{eliminates position } (3,3)):$$

$$\rightarrow \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 3 & -6 & 1 \\ 0 & 0 & 0 & -2 & -5/3 \end{bmatrix}$$

$$\text{Echelon Form} = \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 3 & -6 & 1 \\ 0 & 0 & 0 & -2 & -5/3 \end{bmatrix}$$

Rank = 3 (three nonzero rows — full row rank).

Quick Reference — Linear Algebra**LINEAR ALGEBRA — MASTER FORMULA SHEET**

Key definitions, properties, and results for the exam

Special Matrices

Nilpotent	$A^k = O$
Hermitian	$A = A^* = \overline{A}^t$
Involuntary	$A^2 = I$
Symmetric	$A^t = A$
Skew-Sym.	$A^t = -A$
Idempotent	$A^2 = A$
Orthogonal	$A^t = A^{-1}$
Unitary	$A^* = A^{-1}$

Conjugate Properties

$$\begin{aligned}\overline{A+B} &= \overline{A} + \overline{B} \\ \overline{AB} &= \overline{A} \overline{B} \\ \overline{kA} &= \overline{k} \overline{A} \\ (\overline{A})^t &= \overline{A^t} \\ (A^*)^* &= A \\ (AB)^* &= B^* A^* \\ (A^t)^t &= A \\ (AB)^t &= B^t A^t\end{aligned}$$

Normal Form and Rank

$$\text{Normal Form of rank } r = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \quad PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

Rank = number of nonzero rows in echelon form = number of pivots

$$\text{Rank}(A) + \text{Nullity}(A) = n \quad (\text{number of columns})$$

REF vs RREF

REF (Echelon Form):

- Zero rows at bottom
- Pivot strictly right of row above
- Zeros below each pivot

RREF (Reduced):

- All REF conditions
- Each pivot = 1
- Zeros above AND below pivot

Inverse Formula (2×2)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(A) = ad - bc$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Inverse exists $\iff \det(A) \neq 0$.

For 3×3 : use adjoint method or row reduction on $[A | I]$.

For feedback or to request additional topics:

+92 304 2019543